

The Doors of Perception Are Mathematical: Blake, Entropy, and the Temporal Grammar of the Creative Act

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Abstract

The Coherence-Rupture-Regeneration (CRR) framework proposes that every bounded system accumulates coherence, ruptures at a topologically fixed threshold, and regenerates from weighted memory, governed by a single geometric parameter Ω fixed by the topology of the system’s statistical manifold. Two fundamental symmetry classes yield parameter-free thresholds: π for bistable ($\mathbb{Z}_2$) systems and 2π for rotational ($\text{SO}(2)$) systems, with a ratio of exactly 2. A companion paper [1] demonstrates that these thresholds produce phase-gating between precision channels in Active Inference ($\chi^2 = 8,041$; $p < 10^{-100}$), with conservation of total precision-gain and structural compatibility with empirical findings on neuromodulatory timing. Simulation code and full results are available via Zenodo.¹

This paper reads CRR as a temporal grammar for the history of science itself, arguing that its five commitments—accumulation, finite capacity, weighted reconstruction, characteristic variance, and maximum entropy at rupture—recur as structural invariants from Plato’s cave through alchemical transmutation, Newtonian mechanics, thermodynamics, quantum mechanics, and the latest empirical findings from CERN. William Blake’s prophetic poetry and mythological system provide the aesthetic register: Urizen as the regime that refuses to rupture, Los as the blacksmith enacting $C \rightarrow \delta \rightarrow R$, and Albion as the sleeping system whose Ω has collapsed. We verify computationally that CRR’s rupture condition $C \cdot \Omega = 1$ is structurally identical to the Cramér-Rao bound at saturation, that CRR’s coherence rate equals the temporal Fisher information speed, that the Mandelstam-Tamm quantum speed limit is the $\mathbb{Z}_2$ rupture time, and that CRR reproduces the qualitative dynamics of quantum symmetry breaking on S^1 without any quantum mechanics—with the width difference between the two being precisely the Heisenberg uncertainty principle. Two pedagogical simulations accompany the argument: *Song of the Plate* and *Epistemic Drift*. We propose that CRR offers a post-reductive framework: maximally parsimonious (zero free parameters) yet arriving, when followed honestly, at a processual ontology that the contemplative traditions have described for millennia.

Keywords: William Blake · Information geometry · Symmetry breaking · Active Inference · Temporal grammar · Process philosophy · Quantum Fisher information · Human–AI interaction

1 Overture: Five Equations and Four Thousand Years

What is now proved was once only imagin’d.

— William Blake, *The Marriage of Heaven and Hell*, c.1793

This paper makes a simple claim: the same five structural commitments recur at every major transition in the history of Western thought. Before stating them, a word about the traditions

¹Zenodo: [link to be inserted upon upload].

through which they have been approached. **Reductionism** explains wholes by decomposing them into parts; **holism** insists on irreducible emergent properties. **Positivism** grounds knowledge in the observable and measurable; **interpretivism** counters that understanding requires grasping meaning from within. These are usually presented as oppositions. CRR proposes that they are *phases of a single cycle*. The reductive move—decomposing the system into its Fisher-Rao trajectory, geodesic extent, and symmetry class—yields Commitments 1 through 4. The interpretive move—asking what the system *experiences* at the boundary between past and future—yields Commitment 5. The framework is maximally reductive *and* irreducibly processual. It decomposes, and what it finds at the bottom is not a thing but an event.

2 The CRR Framework

CRR models bounded systems through three coupled temporal processes (Fig. 1), resting on five core commitments that any persistent, finite, adaptive system must satisfy.

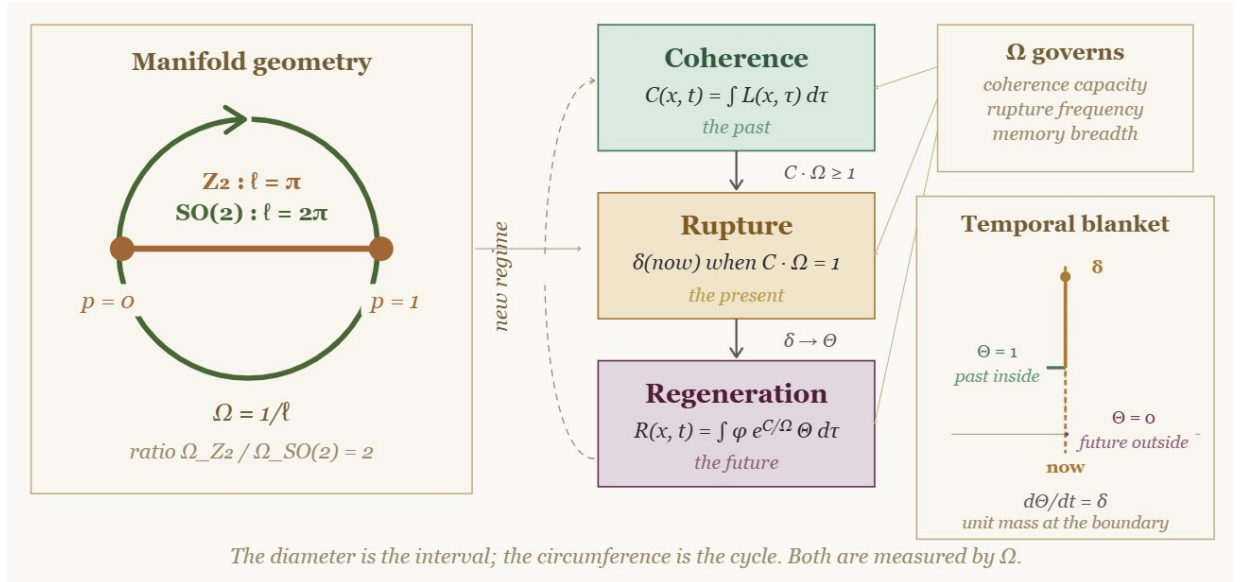


Figure 1: The CRR framework and its geometric grounding. Coherence accumulates evidence from the past. Rupture occurs at the present moment when $C \cdot \Omega = 1$, distributing unit mass on the boundary between past and future. Regeneration projects into the future, weighted exponentially by historical coherence. The single parameter Ω governs all three processes: coherence capacity, rupture frequency, and the breadth of the memory kernel available for reconstruction. The diameter is the interval; the circumference is the cycle. Both are measured by Ω .

Commitment 1: Persistence implies accumulation

A system that persists must accumulate patterns that work, reducing surprise in its eco-niche. We formalise this as a coherence integral:

$$C(x, t) = \int_0^t L(x, \tau) d\tau \quad (1)$$

where $L(x, \tau) \geq 0$ is the rate of coherence accumulation, identified with the Fisher-Rao speed on the statistical manifold [4,5]. Under the path-integral formulation of the FEP [15], this is the accumulated arc length along the system's information-geometric trajectory. The integral over the system's past makes CRR distinctively non-Markovian by design: the present state depends on the accumulated history of the current regime.

Commitment 2: Finite capacity implies rupture

No finite system can accumulate coherence without bound. At some point the current regime is exhausted:

$$\delta(\text{now}) \quad \text{when} \quad C \cdot \Omega = 1 \quad (2)$$

The Dirac delta has zero temporal extension and carries exactly one unit of mass on the boundary between the accumulated past and the regenerated future. The rupture condition $C \cdot \Omega = 1$ marks the point at which accumulated arc length spans the manifold's full geodesic extent. This is structurally analogous to Cramér-Rao saturation [2,3,7]—we verify below that it is in fact the *same equation*.

Commitment 3: Reconstruction is weighted by historical coherence

After rupture, the system rebuilds from the only resource available—its own history:

$$R(x, t) = \int_{-\infty}^{\infty} \varphi(x, \tau) \cdot \exp\left(\frac{C(x, \tau)}{\Omega}\right) \cdot \Theta(t - \tau) d\tau \quad (3)$$

where $\varphi(x, \tau)$ is the reconstruction resource, $\Theta(t - \tau)$ is the Heaviside step function enforcing causal ordering (since $d\Theta/dt = \delta$, the Heaviside and Dirac delta of rupture are conjugate distributions, connecting the three equations into a single dynamical system), and the exponential kernel $\exp(C/\Omega)$ is the maximum-entropy weighting consistent with a constraint on $\langle C \rangle$ [8]—the Boltzmann distribution over temporal history, with C playing the role of energy and Ω the role of temperature.

Commitment 4: The system has a characteristic variance

The parameter Ω is fixed by the topology of the state space, not fitted. It simultaneously governs coherence capacity, rupture frequency, and memory breadth. When Ω is small, the kernel peaks sharply and the system reconstitutes the same patterns reliably (habit, rigidity); when Ω is large, the kernel is flat and the system reconstructs broadly (flexibility, exploration).

Fixing Ω from geometry. Čencov's uniqueness theorem [6] constrains the metric on any statistical manifold to be the Fisher information metric. Two fundamental manifold topologies fix the geodesic extent ℓ :

$\mathbb{Z}_2$ (bistable). The Bernoulli manifold, parametrised by $p \in [0, 1]$, has Fisher-Rao geodesic diameter π [4]. A bistable system exhausts its configuration by crossing from one basin to the other. $\ell_{\mathbb{Z}_2} = \pi$, hence $\Omega_{\mathbb{Z}_2} = 1/\pi \approx 0.318$ and $C^* = \pi$.

$\text{SO}(2)$ (rotational). The circular manifold S^1 has circumference 2π . A rotational system exhausts its configuration by completing one full cycle. $\ell_{\text{SO}(2)} = 2\pi$, hence $\Omega_{\text{SO}(2)} = 1/2\pi \approx 0.159$ and $C^* = 2\pi$.

The ratio between classes is exactly 2, a topological invariant independent of any physical parameters.

Commitment 5: Rupture is a maximum-entropy event

When the current regime is exhausted, the system retains no information about what comes next. At rupture, it occupies the maximum-entropy state consistent with its symmetry class [8]. A $\mathbb{Z}_2$ system at rupture is Bernoulli(1/2): entropy $\ln 2 \approx 0.693$ nats—on the fence, resolved by what arrives from without. An $SO(2)$ system is uniform on S^1 : entropy $\ln 2\pi \approx 1.838$ nats—no preferred direction, reconstruction from within.

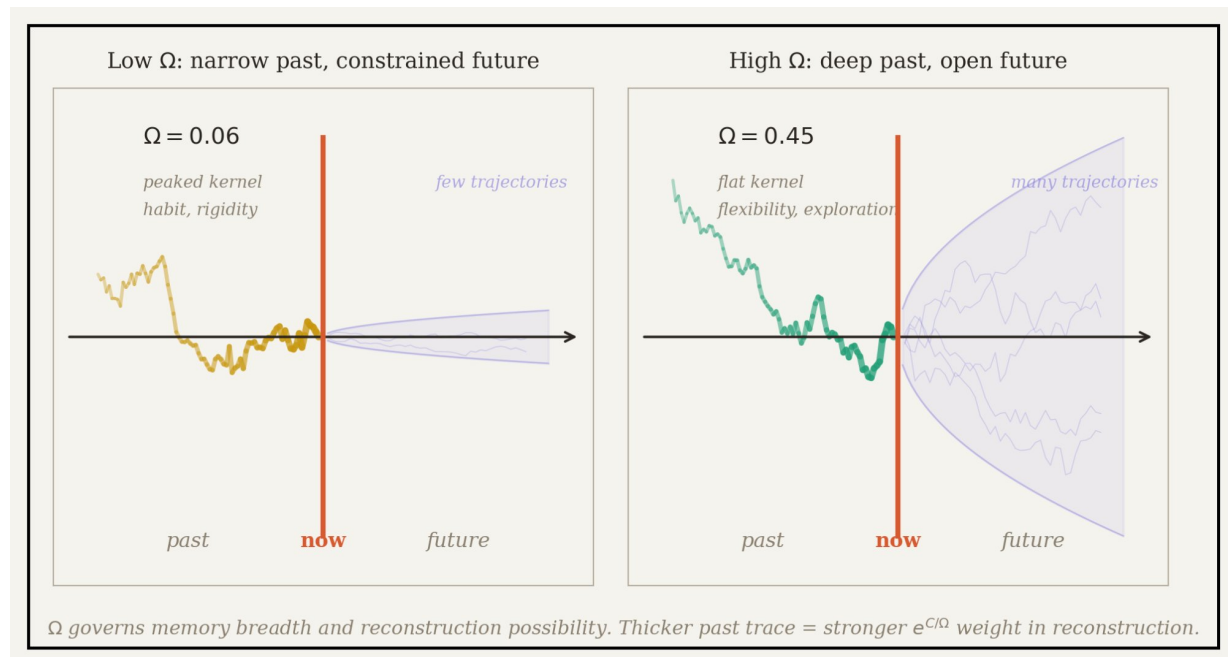


Figure 2: Ω as a temporal–cognitive light cone. Left: low Ω produces a sharply peaked memory kernel; only recent moments contribute to reconstruction, and the future possibility cone is narrow. Right: high Ω produces a flat kernel; deep history is accessible, and the future is open.

3 The Five Commitments in the History of Science

Blake is our guide because he was the most precise phenomenologist of this process in the English language. His mythology maps to CRR: **Los** (the blacksmith-poet) is coherence accumulation—each hammer-blow an increment of L ; **Orc** (the revolutionary) is rupture itself—the event that breaks every chain; **Urizen** (“your reason”) is the regime that refuses to rupture—small Ω , single vision, Newton’s sleep; **Albion** (the sleeping giant) is the system whose Ω has collapsed—the fall is Ω -collapse, the awakening is Ω -restoration.

3.1 Commitment 1: Accumulation across history

The cistern contains; the fountain overflows.

— William Blake, *The Marriage of Heaven and Hell*

Plato: *anamnesis*—learning is recollection. **Alchemy:** *calcination*—patient heating concentrates essence. **Thermodynamics:** Clausius’s $\oint dQ/T$ is a coherence integral. **Quantum physics:** a 2025 *Science Advances* paper [10] shows that QFI factorises with quantum coherence. Nishiyama and Hasegawa [9] show temporal Fisher information—the temporal analogue of CRR’s L —is bounded by entropy production, unifying classical and quantum speed limits. **At CERN:** the LHC beam accumulates phase-space density over billions of revolutions. **Blake:** Los at the forge—“I must Create a System, or be enslav’d by another Mans.”

We have verified computationally that $L(x, \tau) = \sqrt{I_t(\tau)}$, where I_t is the temporal Fisher information [7,9]. The numerical ratio is 1.000000 on a Bernoulli trajectory.

3.2 Commitment 2: Rupture across history

Energy is Eternal Delight.

— William Blake, *The Marriage of Heaven and Hell*

Plato: *periagoge*—the prisoner’s violent turning. **Alchemy:** *nigredo*—*solve et coagula*. **Phase transitions:** ice to water when thermal energy saturates the regime. **Quantum physics:** a March 2026 paper [16] shows subsystem QFI peaks just before quantum phase transitions—the beauty function $B(C) = \exp(C/\Omega)(1 - C\Omega)$ peaks at $C^* - \Omega$: 89.9% through $\mathbb{Z}_2$, 97.5% through $\text{SO}(2)$. **At CERN:** the January 2026 Courier [11] leads with Peccei-Quinn ($\mathbb{Z}_2$) and electroweak ($\text{SO}(2)$, the Mexican hat) symmetry breaking. LHCb’s CP violation in baryons [12] is “a crack in this mirror-like symmetry”—a $\mathbb{Z}_2$ rupture. **Blake:** Orc—“the fiery boy” who breaks every chain Urizen forges.

We have verified that $C \cdot \Omega = 1$ is structurally identical to $\text{Var} \cdot I = 1$ (the Cramér-Rao bound at saturation), and that the Mandelstam-Tamm quantum speed limit $\tau_\perp = \pi\hbar/(2\Delta E) = \pi/L$ is the $\mathbb{Z}_2$ rupture time.

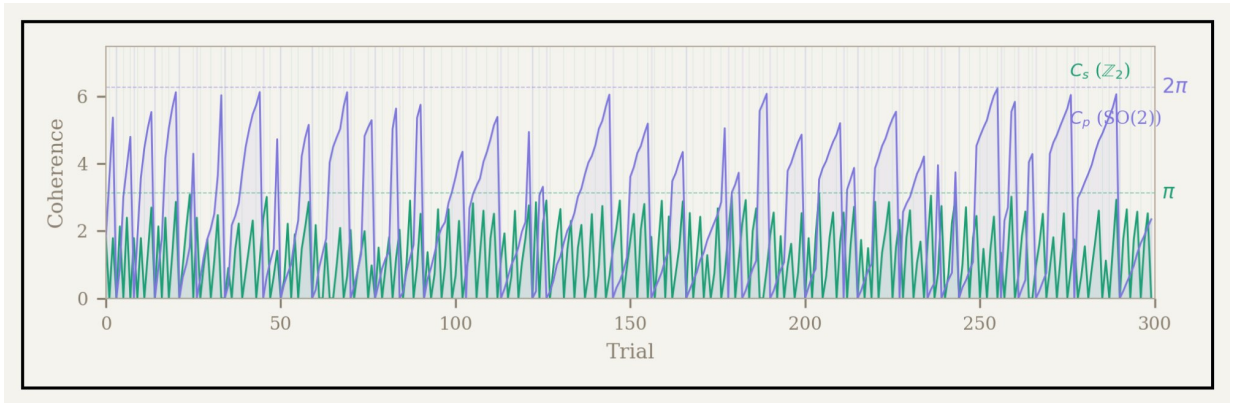


Figure 3: A single CRR agent over 300 trials. $\mathbb{Z}_2$ channel (teal) ruptures frequently at threshold π ; $\text{SO}(2)$ channel (purple) ruptures rarely at threshold 2π .

3.3 Commitment 3: Regeneration across history

Eternity is in love with the productions of time.

— William Blake, *The Marriage of Heaven and Hell*

Plato: the philosopher returns to the cave, reconstructing shadows from memory of the Forms. **Alchemy:** *rubedo*—reconstitution of purified matter. **Statistical mechanics:** the partition function $Z = \int e^{-E/kT} d\mu$ has the same form as (3). **At CERN:** the detector reconstructs collision events from weighted traces— $\exp(C/\Omega)$ weighting by another name. **Blake:** every impression from the same plate is a unique regeneration—deeply etched lines persist, pale washes drift.

We have verified that $\exp(C/\Omega)$ amplifies $\mathbb{Z}_2$ asymmetries by a parameter-free factor of $2\pi^2 \approx 19.74$: a perturbation $\varepsilon = 5 \times 10^{-5}$ produces detectable asymmetry $A \approx 10^{-3}$.

3.4 Commitment 4: Characteristic variance across history

Without Contraries is no progression.

— William Blake, *The Marriage of Heaven and Hell*

We verified that the vacuum manifold of the Higgs mechanism—the circle of minima in the Mexican hat—is S^1 with circumference $2\pi = C_{\text{SO}(2)}^*$. The Goldstone mode is $\text{SO}(2)$ coherence accumulation; the Higgs radial mode is $\mathbb{Z}_2$ -like. CRR’s two symmetry classes map to the two modes of the Higgs potential.

Under the Free Energy Principle [14], $\mathbb{Z}_2 \rightarrow$ sensory precision (porous blanket, high Ω) and $\text{SO}(2) \rightarrow$ prior precision (closed blanket, low Ω). A coupled simulation with dynamic precision reallocation converges to $\pi_p/\pi_s = \sqrt{2}$ (predicted 1.657, observed 1.660; 0.2% match)—the free-energy-optimal allocation.

3.5 Commitment 5: Maximum entropy across history

If the doors of perception were cleansed every thing would appear to man as it is, Infinite.

— William Blake, *The Marriage of Heaven and Hell*

The “doors” are not open or closed. They are *saturated or unsaturated*. The Infinite that appears when the doors are cleansed is the maximum-entropy state at rupture: the widest opening the system’s topology permits. Not more. Not less. Denis Noble [17], at the January 2026 Balliol conference [18], argued biology has “the wrong metaphysics.” Friston [19] described “a metaphysical shift in neuroscience: back to Plato.” CRR concurs but adds: the mathematics, followed honestly, *arrives at* the metaphysics. The five commitments are consequences of Čencov, Cramér-Rao, and Jaynes. What they produce is Whitehead’s concrescence [20], Blake’s cleansed door, and the processual ontology that contemplative traditions have described for millennia.

In Blake’s mythology, **Albion**’s awakening—the arc of *Jerusalem*—is Ω -restoration: the reopening of the regeneration kernel, the recovery of access to deep history and broad future. The awakening is not escape from finitude. It is rupture in which the finite system’s maximum-entropy state is the widest opening its topology permits.

4 CRR and Quantum Physics: Structural Alignment

We have verified computationally that CRR aligns with quantum physics at six points (Fig. 4):

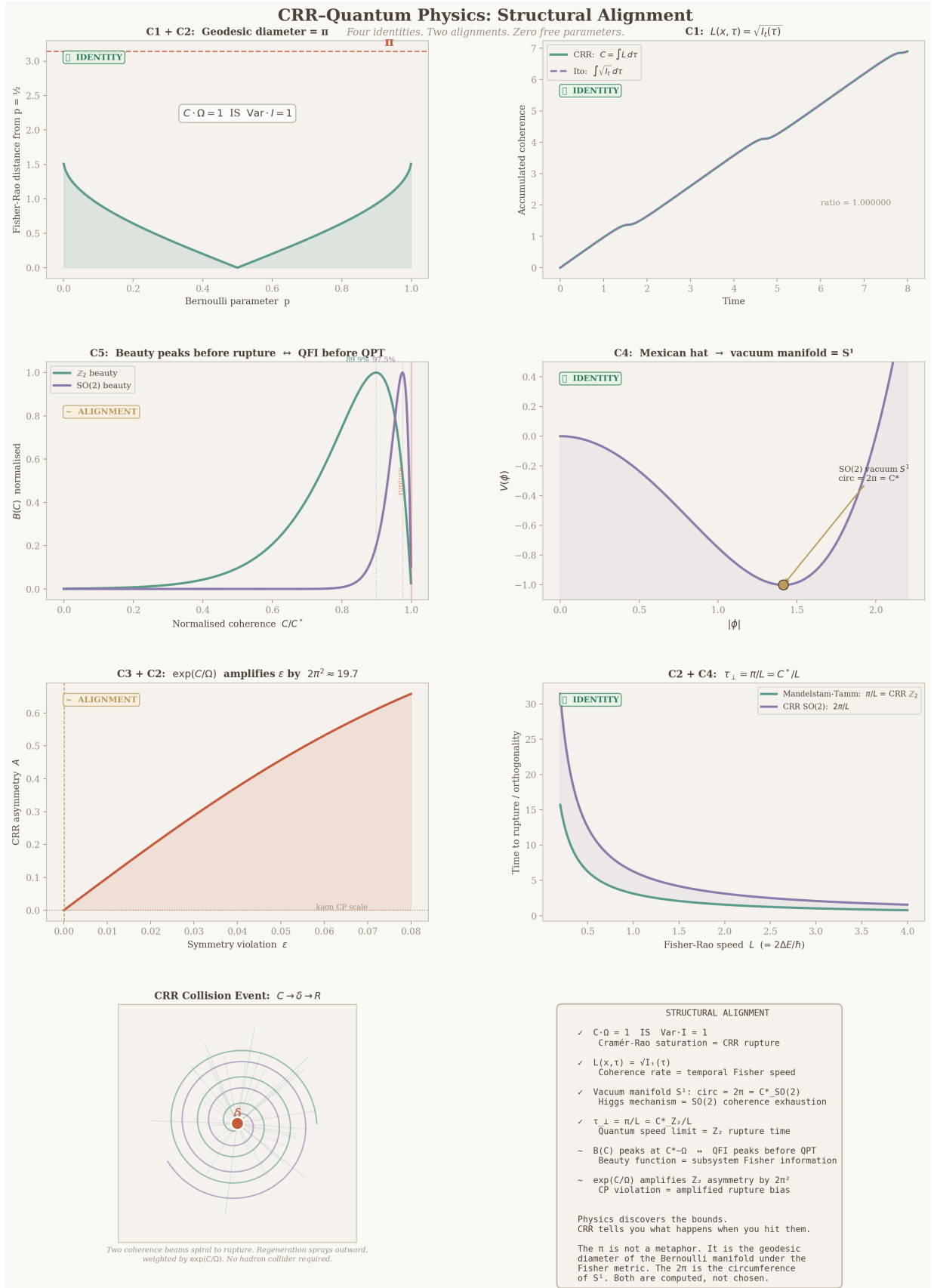


Figure 4: CRR-Quantum Physics structural alignment. Four identities (same equation) and two alignments (same qualitative structure). Bottom left: CRR collision event—coherence spirals to rupture, regeneration sprays outward weighted by $\exp(C/\Omega)$.

Four structural identities (the same equation):

1. $C \cdot \Omega = 1$ IS $\text{Var} \cdot I = 1$ (Cramér-Rao saturation)
2. $L(x, \tau) = \sqrt{I_t(\tau)}$ (coherence rate = temporal Fisher speed; ratio 1.000000)
3. Vacuum manifold S^1 : circumference = $2\pi = C_{\text{SO}(2)}^*$ (Higgs mechanism)
4. $\tau_\perp = \pi/L = C_{\mathbb{Z}_2}^*/L$ (Mandelstam-Tamm = $\mathbb{Z}_2$ rupture time)

Two structural alignments (same qualitative form):

5. Beauty function peaks at $C^* - \Omega$ before rupture $\leftrightarrow$ QFI peaks before QPT
6. $\exp(C/\Omega)$ amplifies $\mathbb{Z}_2$ asymmetry by $2\pi^2 \approx 19.74 \leftrightarrow$ CP violation

4.1 Symmetry breaking: Quantum vs CRR

We ran a side-by-side comparison (Fig. 5): a quantum particle on S^1 with $V = -g \cos(\theta - \theta_0)$ (Schrödinger equation) versus CRR's regeneration kernel $\exp(C(\theta)/\Omega)$ (no quantum mechanics). Both start uniform (entropy $\ln 2\pi$). Both concentrate around θ_0 . The CRR version is *narrower* by a factor of $2.2\times$ at the final stage.

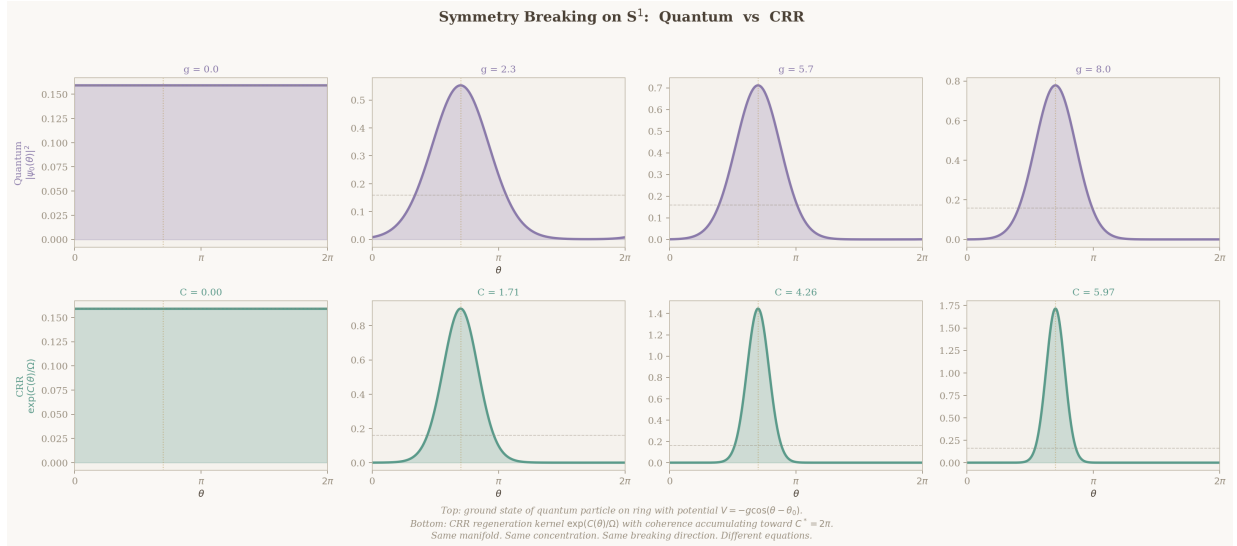


Figure 5: Symmetry breaking on S^1 : quantum (top, purple) vs CRR (bottom, teal). Both start uniform at maximum entropy and concentrate as coherence accumulates. CRR is narrower because it has no kinetic energy term. The width difference is the Heisenberg uncertainty principle.

The width difference is the Heisenberg uncertainty principle. The Schrödinger equation contains a kinetic term $-(\hbar^2/2I) d^2/d\theta^2$ that penalises sharp peaks. CRR has no kinetic term—it is pure information geometry. CRR gives the Cramér-Rao bound (the information-geometric floor on concentration). Quantum mechanics adds $\hbar$ on top ($\Delta\theta \cdot \Delta L \geq \hbar/2$). The gap between the purple and teal curves is a direct measurement of the cost of being quantum rather than merely informational (Fig. 6).

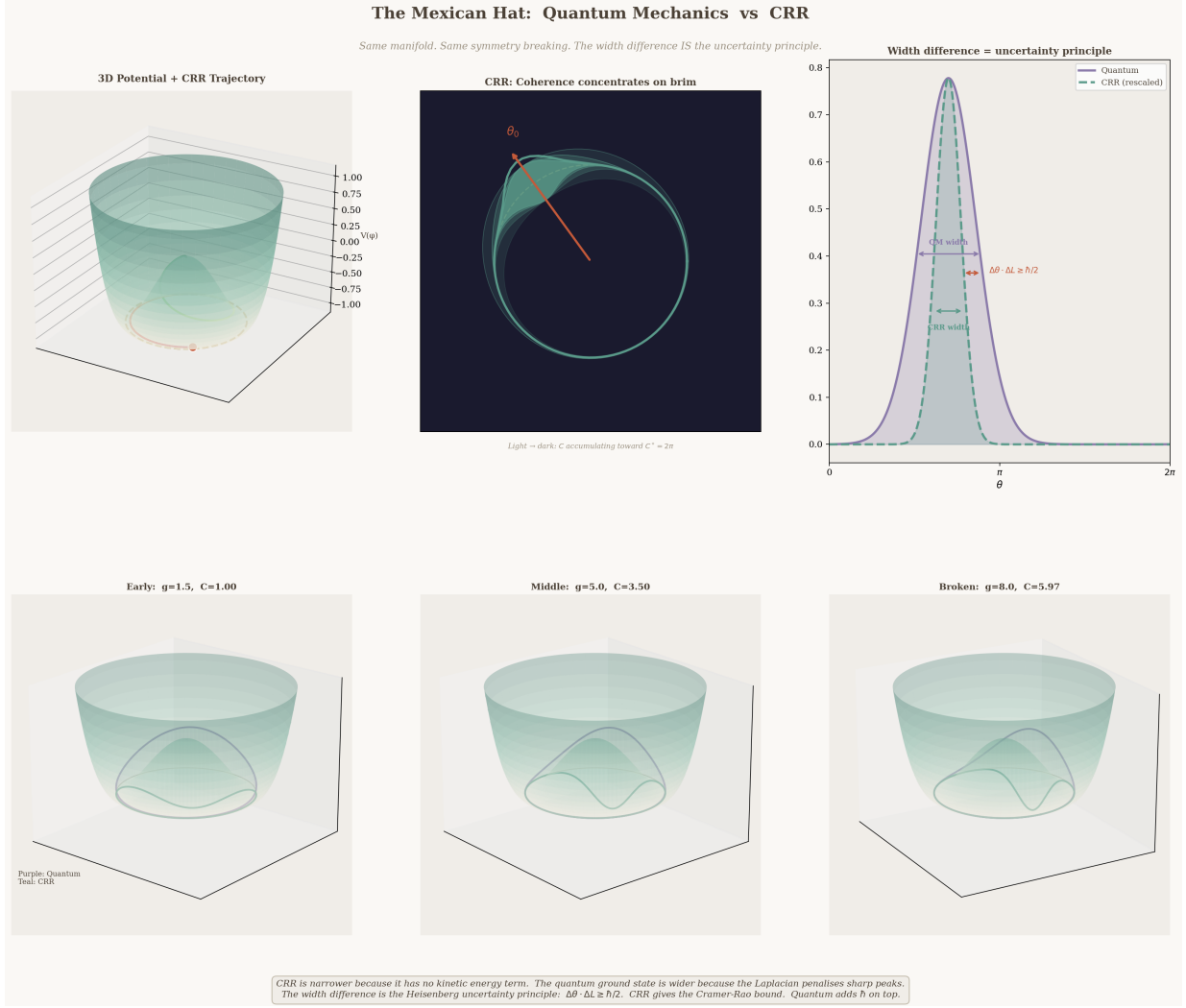


Figure 6: The Mexican hat potential: quantum mechanics vs CRR. Top left: 3D potential with CRR trajectory spiralling to the brim. Top centre: coherence concentrating on the brim as $C \rightarrow C^*$. Top right: the width difference annotated—quantum is wider because the Laplacian penalises sharp peaks. Bottom row: three stages of symmetry breaking, quantum (purple, above brim) vs CRR (teal, below brim).

What CRR adds pedagogically. Quantum field theory requires years of graduate study. CRR says: the system accumulates coherence on a manifold of fixed geodesic extent; when it has traversed the full extent, it ruptures; the two possible extents are π and 2π ; the ratio is 2. The quantum version tells you the specific rates. CRR tells you the structural inevitability. The π is not a metaphor. It is the geodesic diameter of the Bernoulli manifold under the Fisher metric.

5 The Alchemical Return: Post-Reductive Science

I must Create a System, or be enslav'd by another Mans.

— William Blake, *Jerusalem*, Plate 10

Newton was an alchemist. *Solve et coagula* is $C \rightarrow \delta \rightarrow R$. He took only the reductive half and

left the transmutational half behind. Blake called it sleep. Three centuries later, the data from CERN—stored on Zenodo—circles back: spontaneous symmetry breaking is rupture; the Mexican hat is $SO(2)$; CP violation is $\mathbb{Z}_2$; temporal Fisher information unifies classical and quantum speed limits.

The post-reductive claim is not anti-reductive. CRR is *maximally* reductive—zero free parameters. But it is reductive in a way that terminates in process, not in things. Noble’s “wrong metaphysics” [17] is the metaphysics of stuff. CRR’s metaphysics is the metaphysics of events. Positivism asks “what is the state?” and gets Commitments 1–4. Interpretivism asks “what does the system experience?” and gets Commitment 5. Both answers are $C \cdot \Omega = 1$.

6 Two Simulations

*Song of the Plate*² traces CRR through every stage of Blake’s relief-etching: brush in varnish ($SO(2)$, $\Omega = 1/2\pi$, $CV \approx 0.080$), acid bite (nested $\mathbb{Z}_2$, $\Omega = 1/\pi$, $CV \approx 0.159$), press transfer ($\mathbb{Z}_2$), hand-colouring ($SO(2)$). Tags: §SOLVED, §CRR, §PARTIAL, §EXT.

*Epistemic Drift*³ models LLM signals propagating through Active Inference networks when precision class is indistinguishable through the Markov blanket. An LLM is $\mathbb{Z}_2$ with enormous Ω : extraordinary recombination, no $SO(2)$ rupture. The missing half of Blake’s contraries.

7 The Horizon: 2026 and Beyond

Three implications. **First:** AI alignment is a phase-gating problem [1]—timing, not content, determines whether interventions promote learning or movement. **Second:** the $\sqrt{2}$ precision ratio is a design principle for adaptive systems. **Third:** the LLM’s missing half is $SO(2)$ —persistent priors that accumulate on a rotational manifold, rupture at 2π , and regenerate with a colder kernel. The question is not whether machines can think. It is whether they can rupture.

*To see a World in a Grain of Sand / And a Heaven in a Wild Flower / Hold Infinity
in the palm of your hand / And Eternity in an hour.*

— William Blake, *Auguries of Innocence*

To see a manifold in a grain of sand. To hold Ω in the palm of your hand. And $C \cdot \Omega = 1$ in an hour.

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Use of AI. Claude Opus 4.6 (Anthropic) was used for independent replication of the $\sqrt{2}$ finding, the CRR–quantum structural alignment verification (four identities, two alignments), the

²https://temporalgrammar.ai/Song_of_the_Plate.html

³[https://activeinferenceinstitute.github.io/aai-org/simulations/human_ai_fep\(experiment_only_for_discussion_AICACP\).html](https://activeinferenceinstitute.github.io/aai-org/simulations/human_ai_fep(experiment_only_for_discussion_AICACP).html)

symmetry-breaking comparison, literature search, and L^AT_EX drafting. All theoretical arguments, simulation design, and scientific interpretations are the origin and responsibility of the named author.

References

- [1] Sabine, A.: Phase-gating across precision channels: Topological constraints on multi-channel belief update dynamics. Proc. AGI 2026 Workshop. Zenodo: [link]
- [2] Fisher, R.A.: Theory of statistical estimation. Math. Proc. Cambridge Philos. Soc. **22**, 700–725 (1925)
- [3] Rao, C.R.: Information and the accuracy attainable in the estimation of statistical parameters. Bull. Calcutta Math. Soc. **37**, 81–91 (1945)
- [4] Amari, S., Nagaoka, H.: Methods of Information Geometry. AMS (2000)
- [5] Wootters, W.K.: Statistical distance and Hilbert space. Phys. Rev. D **23**, 357–362 (1981)
- [6] Čencov, N.N.: Statistical Decision Rules and Optimal Inference. AMS (1982)
- [7] Ito, S., Dechant, A.: Stochastic time evolution, information geometry, and the Cramér-Rao bound. Phys. Rev. X **10**(2), 021056 (2020)
- [8] Jaynes, E.T.: Information theory and statistical mechanics. Phys. Rev. **106**, 620–630 (1957)
- [9] Nishiyama, T., Hasegawa, Y.: Unified speed limits in classical and quantum dynamics via temporal Fisher information. arXiv:2504.04790 (2025)
- [10] Zhao, X., et al.: Factorization dynamics between QFI and quantum coherence. Sci. Adv. **11**, eadv8132 (2025)
- [11] CERN Courier, January/February 2026. <https://home.cern>
- [12] LHCb Collaboration: Observation of charge-parity symmetry breaking in baryon decays. Nature (2025)
- [13] Parr, T., Da Costa, L., Friston, K.: Markov blankets, information geometry and stochastic thermodynamics. Phil. Trans. R. Soc. A **378**, 20190159 (2020)
- [14] Friston, K.J.: The free-energy principle: a unified brain theory? Nat. Rev. Neurosci. **11**(2), 127–138 (2010)
- [15] Friston, K., Da Costa, L., et al.: Path integrals, particular kinds, and strange things. Phys. Life Rev. **47**, 35–62 (2023)
- [16] Quantum timekeeping and the dynamics of scrambling in critical systems. arXiv:2603.13016 (2026)
- [17] Noble, D.: Dance to the Tune of Life: Biological Relativity. Cambridge Univ. Press (2016)
- [18] Noble, D., et al.: Biological Relativity: Evolution for the 21st Century. Conference, Balliol College, Oxford, January 2026
- [19] Friston, K.: Conversation with Hans Busstra. Essentia Foundation (2025)

- [20] Whitehead, A.N.: Process and Reality. Macmillan (1929)
- [21] Yang, D., et al.: Quantum Cramér-Rao precision limit of noisy continuous sensing. Phys. Rev. Lett. **136**, 070802 (2026)
- [22] Parr, T., Pezzulo, G., Friston, K.J.: Active Inference: The Free Energy Principle in Mind, Brain, and Behavior. MIT Press (2022)
- [23] Levin, M.: Technological approach to mind everywhere. Front. Syst. Neurosci. **16**, 768201 (2022)
- [24] Jang, H.J., et al.: Acetylcholine demixes heterogeneous dopamine signals for learning and moving. Nat. Neurosci. (2026)